

ML Notes

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1 Mathematical Concepts

1.1 Eigenvectors and Eigenvalues

Let $A = [a_{ij}]_{n \times n}$. Consider the homogeneous system of equation,

$$AX = \lambda X$$

Or,

$$(A - \lambda I) = 0$$

Which gives the *characteristic equation*. Upon solving, the values λ are found, which are the eigen values.

Put these λ in the first equation and solve for X to find the corresponding eigenvector.

1.2 Singular Value Decomposition (SVD)

It is a way of computing U , Σ , V , where $A = U\Sigma V^T$

Step 1 - V :

- Find the value of $A^T A$
- Find eigen values & eigen vectors of $A^T A$. Write the eigen values in descending order ($\lambda_1 > \lambda_2 \dots$)
- Write the corresponding normalized eigen vectors in columns of a matrix
- This matrix is V .

Step 2 - Σ :

- Order of Σ = Order of A .
- $\sigma_1 = \sqrt{\lambda_1}$ and $\sigma_2 = \sqrt{\lambda_2}$.
- Rank of Σ = Number of non zero eigen values
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$$\Sigma = \begin{bmatrix} \Sigma_1 & 0 \\ 0 & 0 \end{bmatrix}$$

Where, Σ_1 is a diagonal matrix whose diagonal entries are σ_1 and σ_2 . The zeros are added around Σ_1 to make the Order of Σ = Order of A .

Step 3 - U:

- Find AA^T .
- Find eigenvalues ($\lambda_1 > \lambda_2 \dots$) and eigenvectors (normalized) of AA^T .
- U is the matrix found by appending the eigen vectors one after the other.

2 Polars

Two important things,

- Expressions
- Contexts – Four most common contexts are,
 1. `select`
 2. `with_columns`
 3. `filter`
 4. `group_by` – rows are grouped according to the unique values of the grouping expressions